Practical no. 6B

Aim: Create a PWA that displays a custom offline page when the user is not connected to the internet.

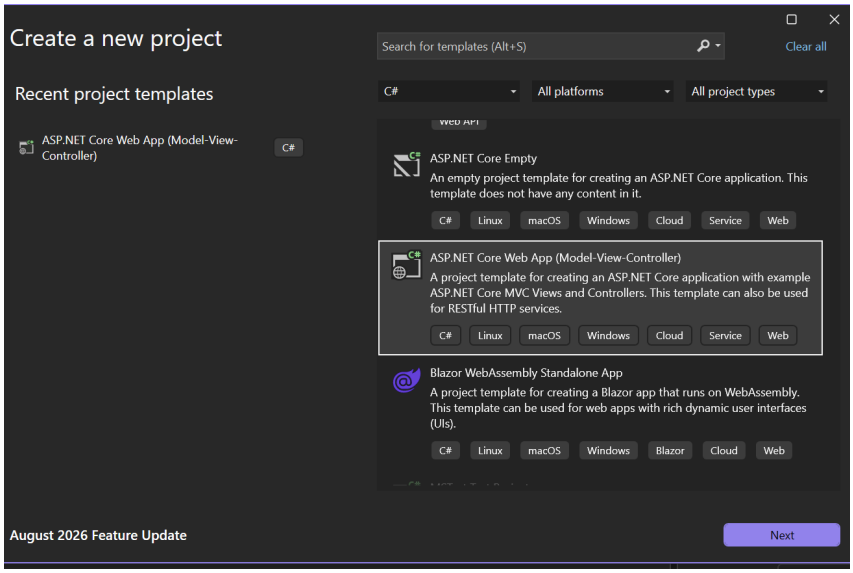
Software Required:

Visual Studio 2026

Procedure:

STEP 1: CREATE NEW PROJECT

1. Open Visual Studio 2026.
2. Select Create a new project.
3. Search for ASP.NET Core Web App (Model-View-Controller).
4. Click Next.
5. Set the Project Name to Practical No. 6b.
6. Select a suitable location and click Next.
7. Select the installed/latest .NET version and keep the default options.
8. Click Create.



STEP 2: CREATE HOME PAGE

Open Views → Home → Index.cshtml and replace its contents with:

```
@{
    ViewData["Title"] = "PWA Offline Demo";
}

<div style="text-align:center; margin-top:50px;">
```

```
    <h1>Welcome to My PWA</h1>

    <p>
        This is a Progressive Web Application.
    </p>

    <p>
        The application provides a custom page when
        the Internet connection is unavailable.
    </p>

</div>
<script>

    if ("serviceWorker" in navigator) {

        window.addEventListener("load", function () {

            navigator.serviceWorker
                .register("/service-worker.js")
                .then(function () {
                    console.log("Service Worker registered successfully.");
                })
                .catch(function (error) {
                    console.log("Service Worker registration failed:", error);
                });

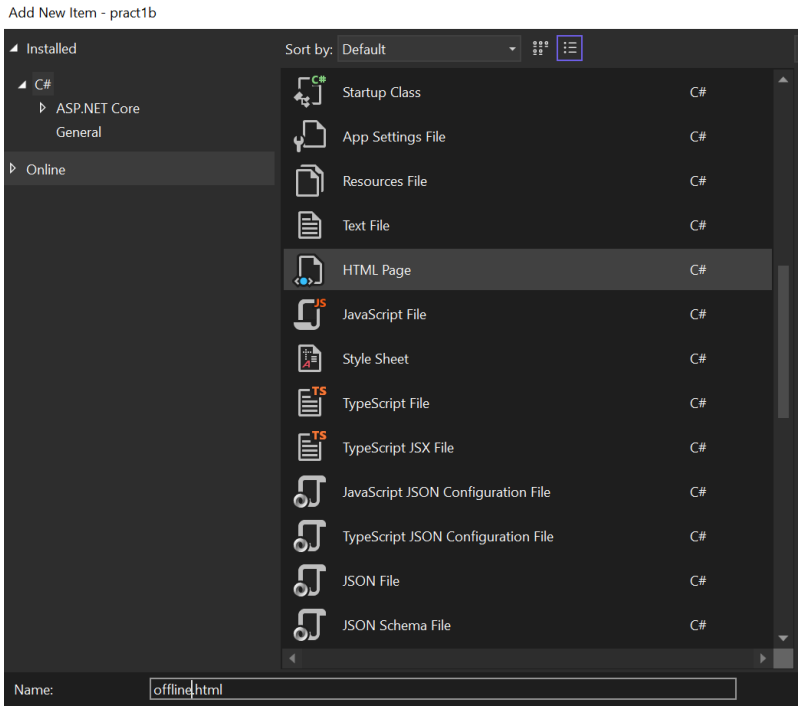
        });

    }

</script>
```

Step 3: Create the Custom Offline Page

1. Right Click → **wwwroot** folder
2. Click Add → New item
3. Select html page.
4. Enter File Name: **offline.html**
5. Click Add.



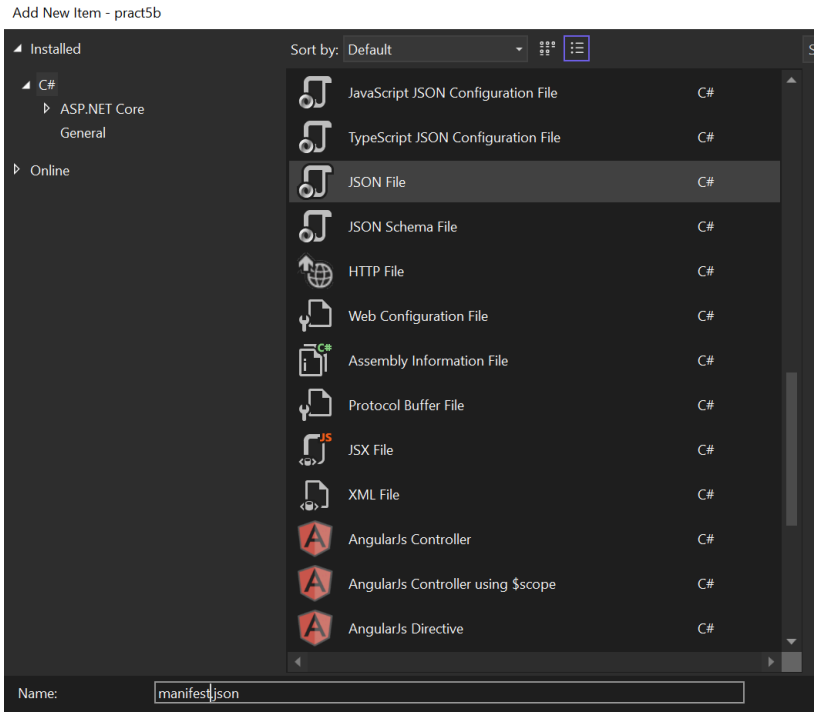
Write the following code.

offline.html

```
<!DOCTYPE html>
<html>
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>You are Offline</title>
</head>
<body style="text-align:center; font-family:Arial; padding:60px;">
   <h1> You are Offline</h1>
   <p>It looks like you are not connected to the Internet.</p>
   <p>Please check your Internet connection and try again.</p>
   <button onclick="location.reload()">Try Again</button>
</body>
</html>
```

Step 4: Create manifest.json

1. Right Click → **wwwroot** folder
2. Click Add → New item
3. Select json file.
4. Enter File Name: **manifest.json**
5. Click Add.



Enter:

```
{
  "name": "PWA Offline Demo",
  "short_name": "PWA Demo",
  "start_url": "/",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#007bff"
}
```

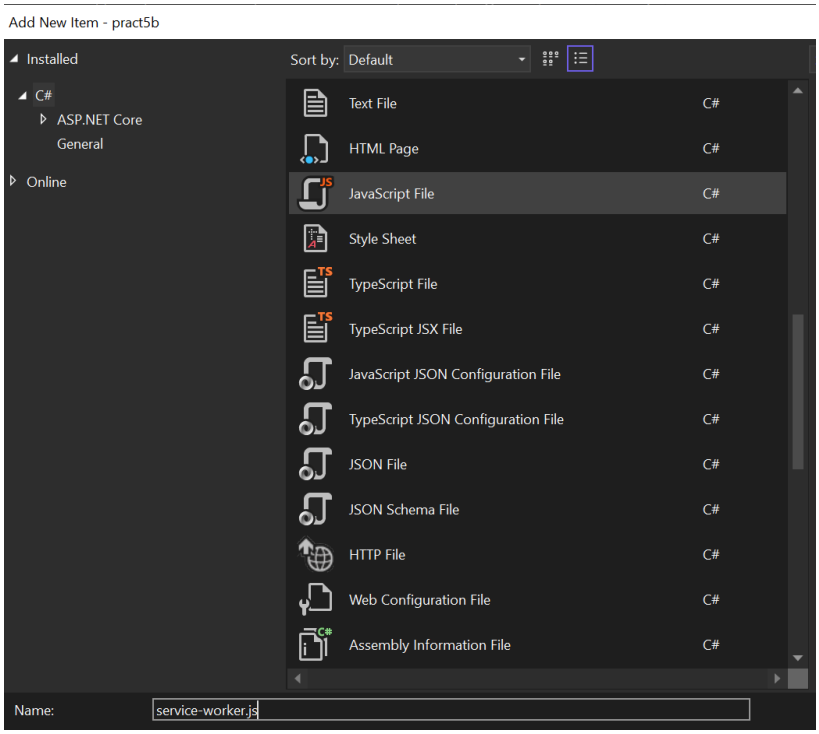
Step 4 : Add the Manifest
Open Views → Shared → _Layout.cshtml.

Inside the <head> section, add:

```
<link rel="manifest" href="/manifest.json">
<meta name="theme-color" content="#007bff">
```

Step 8: Create Service Worker

1. Right Click → **wwwroot** folder
2. Click Add → New item
3. Select JavaScript file.
4. Enter File Name: **service-worker.js**
5. Click Add.



Write the following code in the service-worker.js file.

service-worker.js

```
const CACHE_NAME = "pwa-cache-v1";

const FILES_TO_CACHE = [
    "/",
    "/offline.html",
    "/manifest.json"
];

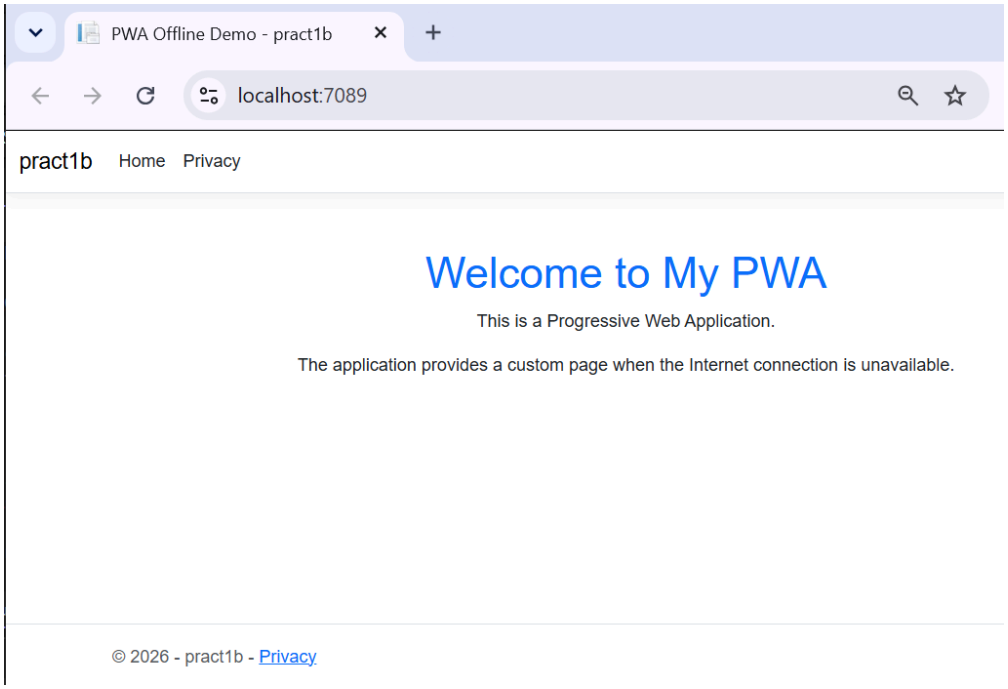
self.addEventListener("install", function (event) {
    event.waitUntil(
        caches.open(CACHE_NAME)
            .then(function (cache) {
                return cache.addAll(FILES_TO_CACHE);
            })
    );
});

self.addEventListener("fetch", function (event) {
    event.respondWith(
        fetch(event.request)
            .catch(function () {
                return caches.match("/offline.html");
            })
```

```
    );
});
```

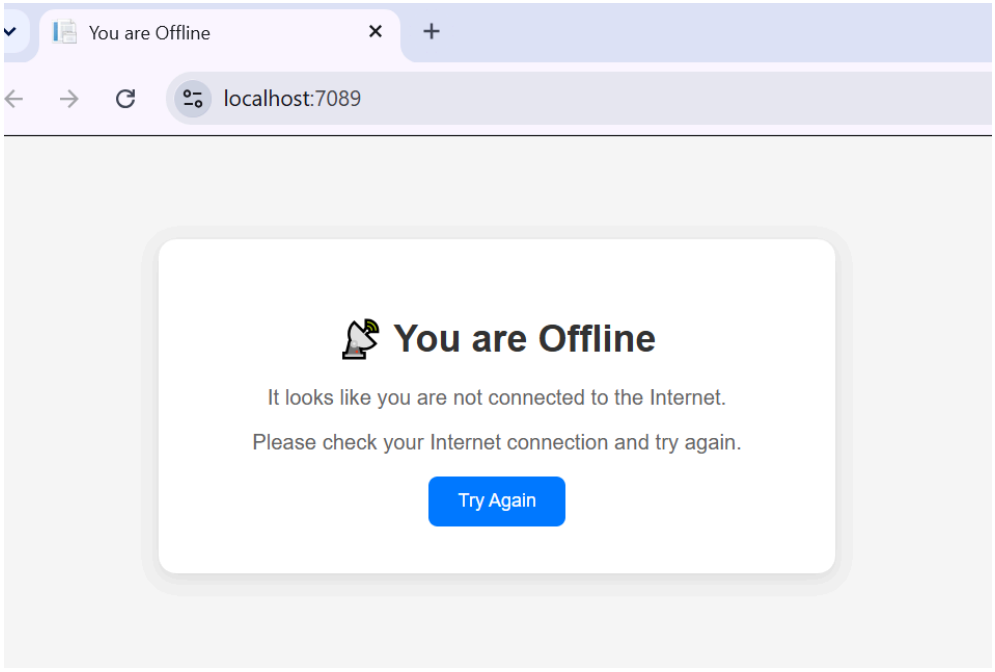
Step 9: Run the Application

Press F5



Step 10: Test Offline Mode

1. Press: F12
2. Open: Network Tab
3. Select: Offline
4. Refresh page.


You are Offline
localhost:7089
You are Offline
It looks like you are not connected to the Internet.
Please check your Internet connection and try again.
Try Again